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EPIDEMIOLOGY

Modeling the transmission mitigation impact of testing for infectious diseases

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A fundamental question of any program focused on the testing and timely diagnosis of a communicable disease is its effectiveness in reducing transmission. Here, we introduce testing effectiveness (TE)—the fraction by which testing and post-diagnosis isolation reduce transmission at the population scale—and a model that incorporates test specifications and usage, within-host pathogen dynamics, and human behaviors to estimate TE. Using TE to guide recommendations, we show that today's rapid diagnostics should be used immediately upon symptom onset to control influenza A and respiratory syncytial virus but delayed by up to two days to control omicron-era severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Furthermore, while rapid tests are superior to reverse transcription quantitative polymerase chain reaction (RT-qPCR) to control founder-strain SARS-CoV-2, omicron-era changes in viral kinetics and rapid test sensitivity cause a reversal, with higher TE for RT-qPCR despite longer turnaround times. Last, we illustrate the model's flexibility by quantifying trade-offs in the use of post-diagnosis testing to shorten isolation times.

INTRODUCTION

Four years after the emergence of SARS-CoV-2, a new status quo for test usage has emerged. Despite documented successes of routine screening for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) in nursing homes (*1*), college campuses (*2*), and even nations (*3*), regular screening via reverse transcription quantitative polymerase chain reaction (RT-qPCR) or rapid diagnostic tests (RDTs) has given way to elective testing after known exposures or symptom onset, typically with RDTs alone. At the same time, the variety and targets of available diagnostics continues to grow, with numerous available RDTs for respiratory syncytial virus [RSV; (*4*)] and influenza A (*5*), new RDTS for SARS-CoV-2 using exhaled aerosols (*6*), and simultaneous testing for all three viruses via both multiplex RT-qPCR (*7*) and rapid antigen lateral flow "triple tests" (*8*). Rapid diagnostics have proved valuable for a broader set of pathogens too, including HIV (*9*) and measles (*10*), as well as *Plasmodium falciparum*, with sufficient impact that rapid diagnosis and treatment have even selected for RDT-escape mutations among *P. falciparum* parasites (*11*).

Direct empirical estimation of the population-scale impact of testing is difficult and only possible retrospectively, elevating the value of mathematical modeling that can predict it from first principles. Mathematical models estimating the impacts of testing on transmission (*2*, *12*–*19*) and treatment (*20*, *21*) proved useful in guiding policy and recommendations for SARS-CoV-2, building in many ways on a broad set of earlier efforts to estimate transmission reduction, clinical impact, and cost effectiveness for routine HIV screening [e.g., (*22*, *23*)]. However, these successes have been restricted to a limited number of pathogens and either routine screening or risk group–based testing, highlighting the need for more flexible modeling to accommodate an increasing array of diagnostic tests for a growing set of pathogens, used electively after known exposures or symptom onset. Moreover, because testing guidelines are only as effective as human behaviors allow them to be, it would be valuable for models to incorporate key behaviors such as imperfect participation (*9*) and compliance (*24*) and imperfect adherence to post-diagnosis isolation (*25*).

Here, we fill this gap by introducing a more general mathematical model for testing without restricting analysis to a single pathogen, test, testing pattern, or set of behaviors. Our focus is to estimate (i) the extent to which testing reduces the risk of transmission for infected individuals, (ii) the distribution of diagnosis times and the probability that individuals are diagnosed at all, and (iii) the costs of test consumption and isolation days corresponding to these diagnosis and transmission mitigation benefits. Together, this model places various intuitions about disease mitigation via testing on firm quantitative ground and exposes important testing-associated variables and behaviors to in silico experimentation and optimization.

To demonstrate how one might use our model, we apply it to the study of the respiratory pathogens RSV, influenza A, and SARS-CoV-2. We first analyze differences between pathogens and between testing strategies by asking whether a single testing strategy might be optimal for all three focal pathogens. Second, we ask when one should test after respiratory symptoms appear, acknowledging that supplies may be limited and infection may not yet be detectable at symptom onset. Third, we reevaluate the trade-offs between test sensitivity and turnaround time (TAT) for SARS-CoV-2 in light of shifts in viral kinetics and symptom onset time caused by immune experience and new variants. Last, we compare the costs and benefits of fixed duration and test-to-exit isolation guidelines using the model's cost estimates for test consumption and days spent in isolation. In this way, this paper has twin goals: to first introduce a flexible mathematical model for testing and then demonstrate its value in the service of relevant scientific and policy questions.

RESULTS

A model for TE

To examine the impact of testing on community transmission, we developed a probabilistic model that integrates four key elements: (i) the properties of a particular diagnostic test; (ii) a strategy for its

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administration; (iii) the time-varying profiles of infectiousness, symptoms, and detectability over the course of an infection; and (iv) the key behaviors of participation (whether or not one refuses to test), compliance (whether or not one takes a recommended test), and isolation length. Given these elements, the model generates a distribution of probable diagnosis times and uses them to compute the expected reduction in the risk of transmission due to testing. By then incorporating heterogeneity between individuals, the model estimates testing effectiveness (TE), the proportion by which a testing program reduces population-level transmission, in expectation, given by the relationship between reproductive numbers R with and without testing,

$$TE = 1 - \frac{R_{\text{testing}}}{R_{\text{no testing}}} \quad \leftrightarrow \quad R_{\text{testing}} = (1 - TE)R_{\text{no testing}} \qquad (1)$$

This definition intentionally mirrors the language of vaccine effectiveness (VE) against onward transmission, that is, the expected reduction in postinfection transmission risk for vaccinated versus unvaccinated individuals (*26*, *27*). However, we name the present quantity testing effectiveness instead of test effectiveness due to the simple observation that the same test, used differently, may have a markedly different impact.

We built our analyses around a simple, common, and computationally efficient model of within-host pathogen kinetics: After some postexposure latent period, the pathogen load grows exponentially at some proliferation rate until reaching a peak and then declines exponentially at some clearance rate. This leads to a piecewise linear model of log pathogen load, which requires only the four parameters of latent, proliferation, and clearance periods, and a peak load, which we draw from distributions estimated from studies of RSV (*28*, *29*), influenza A (*30*–*32*), and SARS-CoV-2 measured for both the founder strain in naive hosts during the pre-vaccine era (*13*, *17*, *33*) and omicron variants in experienced hosts during the post-vaccine era (see table S1) (*33*, *34*). While the results presented here use this simplistic model of viral kinetics, the modeling framework is highly flexible to incorporate more sophisticated alternatives.

We used stochastic realizations from this simple pathogen load model in three ways. First, we assumed that a test taken at time t would return a negative diagnosis if the pathogen load was below the test's limit of detection (LOD) and would return a positive diagnosis with some probability when pathogen load was above the LOD. Because of the importance of delays between a test's sample time and its diagnosis time (*13*), we modeled sample-to-answer delays by returning results after a specified turnaround time (TAT). Second, we took infectiousness to be proportional to the log of pathogen load in excess of an empirically estimated threshold (*35*, *36*), consistent with observations that higher viral loads are associated with more efficient transmission for pathogens including SARS (*37*), SARS-CoV-2 (*24*), influenza (*38*), and RSV (*39*). Alternative relationships between pathogen load and infectiousness are possible (*13*, *36*). Third, we drew symptom onset times relative to the times of peak pathogen load, reflective of the typical manner of reporting in the literature (*38*, *40*). In this way, our model is similar in spirit to the CEPAC (cost-effectiveness of preventing AIDS complications) model, which provides stochastic individual-level realizations of post–HIV-infection dynamics, costs, and outcomes (*41*).

Our model generates its estimates by integrating over the timing of possible symptoms and tests to calculate a distribution of diagnosis times, including the possibility of no diagnosis at all. If receiving a diagnosis, each individual is assumed to isolate thereafter for a specified number of days or until released by a negative test (via a so-called test-to-exit plan), with mitigated infectiousness during isolation (Fig. 1A). By averaging outcomes over the ensemble defined by its random variables, whether by integration or via Monte Carlo, the model produces estimates of the expected infectiousness curves over time, with and without testing (Fig. 1B). The areas under these two curves are proportional to the total transmission

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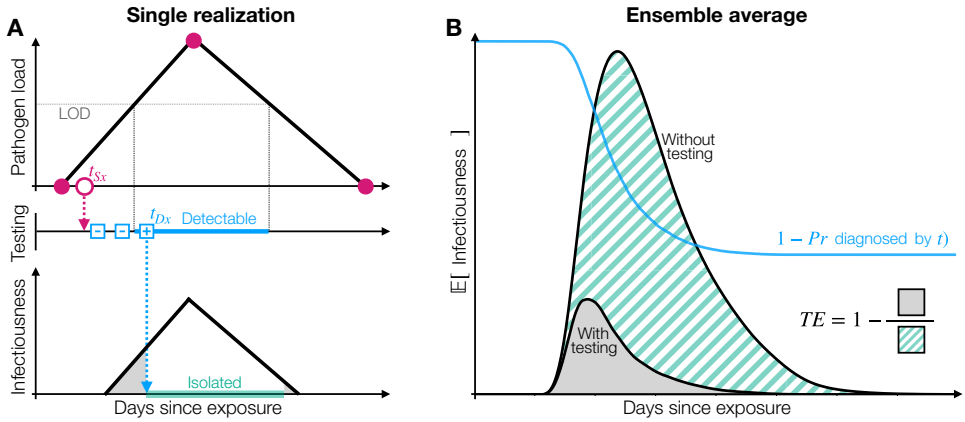

Fig. 1. Model diagram. (**A**) Each realization of the stochastic testing model first draws four control points to specify a piecewise linear model of pathogen load on a logarithmic scale (filled pink circles) and a symptom onset time (t_{Sx}, open pink circle). The realization then draws a set of testing times (open blue squares), which may be triggered by symptoms (pink arrow), triggered by a known exposure (not shown), or ongoing at a particular cadence (not shown). A test taken during the detectable window (blue bar) when pathogen load exceeds the test's LOD (gray line) will return a positive diagnosis with a fixed probability after a specified TAT (not shown). However, not all tests are necessarily taken because of imperfect compliance (not shown). Diagnosis at time t_{Dx} leads to isolation and thus reduced infectiousness (gray). (**B**) The ensemble mean, whether computed through integrals or estimated via Monte Carlo, produces expected infectiousness curves with and without testing. The areas under these curves are proportional to their respective reproductive numbers, enabling estimation of testing effectiveness (TE) (see Eq. 1). The model also computes the proportion of individuals who remain undiagnosed at time t (blue curve), a curve that approaches zero as ascertainment approaches 100%.

potential with and without testing and, thus, their respective reproductive numbers. The model also provides a curve representing the cumulative probability that a randomly chosen infected individual has not yet been diagnosed by some time (Fig. 1B). The long-time limit of this curve is particularly useful because it represents the proportion of the infected population who escape diagnosis entirely. Its complement, the proportion of the infected population to receive a diagnosis at any point, is therefore the ascertainment of the testing scenario, a quantity also called protocol sensitivity in the literature (*19*). A complete mathematical description and details of parameterizations can be found in Materials and Methods.

Testing effectiveness varies by strategy and pathogen

Despite the 2023 end to the World Health Organization's COVID-19 public health emergency (*42*), the burden of COVID-19 and respiratory viruses more broadly remains substantial. In the U.S. alone, an estimated 9 million cases of influenza A caused 100,000 hospitalizations [2021 to 2022 season; (*43*)], and a global estimate of 33 million RSV infections in children under 5 years led to 3.6 million associated hospitalizations and 101,400 associated deaths, the vast majority of which were in low- and middle-income countries [2019; (*44*)]. With the broad expansion of diagnostic testing globally, including at-home RDTs for influenza A, RSV, and SARS-CoV-2, we sought to determine whether a single testing strategy might be optimal for all three common respiratory pathogens.

To examine the potential impacts of testing, we considered three distinct testing behaviors meant to capture both institutional testing strategies and elective testing in response to exposure or symptoms. First, we considered routine weekly RDT screening (TAT = 0; LODs table S1). Given the markedly different sensitivities of different RDT kits and RT-qPCR protocols (*45*, *46*), a representative LOD was chosen for each respiratory virus to investigate general principles of testing (see Materials and Methods). To incorporate the fact that not all policy-prescribed tests are actually taken in practice (*24*), this scenario included compliance of only 50% such that each test was taken or not taken independently with probability one-half. Second, we considered an elective testing scenario in which individuals experiencing symptoms used one RDT per day for two consecutive days following the onset of symptoms ($TAT = 0$; see table S1 for stochastic timing and prevalence of symptoms). Last, we considered an elective testing scenario in which 75% of individuals sought out a single RT-qPCR test (TAT = 2; LODs table S1) between 2 and 7 days after exposure; the other 25% did not participate. For each scenario and each of the three focal respiratory viruses (RSV, influenza A, and SARS-CoV-2), we calculated TE, the timing of diagnoses, and ascertainment—the total fraction of individuals receiving a positive diagnosis—based on 10^5 stochastic realizations of viral load, symptom onset, testing, diagnosis, and isolation. For SARS-CoV-2, we specifically considered within-host dynamics and RDTs associated with omicron variants in immune-experienced hosts.

This analysis demonstrated that a test, how it is used, and the pathogen it targets can interact in potentially complicated ways. For instance, we found that weekly RDT screening with 50% compliance exhibited low TE for all three pathogens, ranging from 13% for omicron-era SARS-CoV-2 down to just 7% for influenza A (Fig. 2A). These low values are driven by the fact that the time windows during which the pathogens are detectable via RDT are almost always shorter than 7 days, meaning that weekly testing missed a considerable number of infections (Fig. 3). These short detection windows are compounded by the assumed 50% compliance, leading to low ascertainment across pathogens (11 to 14%; fig. S1).

In contrast to regular screening, elective testing with the same RDT at symptom onset achieved 38% TE for influenza A but only 21% for RSV (Fig. 2A). This difference in effectiveness primarily reflects differences in the relative timing of symptoms, infectiousness, and detectability: Both influenza A and RSV typically become detectable by RDT within 1 day of symptom onset, but RSV infections are characterized by higher rates of asymptomatic and presymptomatic infectiousness (Fig. 3, A and B). In contrast, TE for elective testing at SARS-CoV-2 symptom onset was middling at 25%, balancing early detection for those who test positive with lower ascertainment as a result of testing too early to be detected (Fig. 3C). Because of

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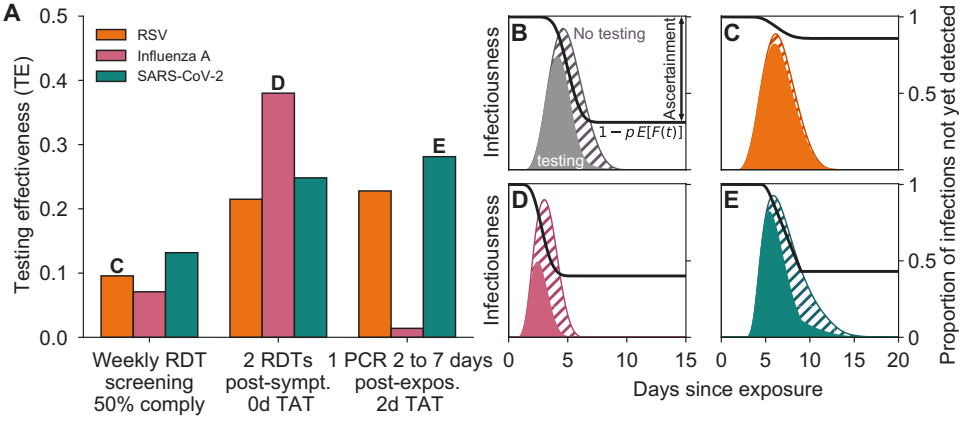

Fig. 2. TE varies considerably by strategy and pathogen. (**A**) Testing effectiveness (TE) is shown for RSV (orange), influenza A (pink), and SARS-CoV-2 omicron in experienced hosts (green) under three testing programs: (i) weekly RDT screening with 50% compliance, (ii) testing with one RDT per day for 2 days starting at symptom onset, and (iii) one RT-qPCR test administered 2 to 7 days after exposure, with 75% participation and 2 days test TAT. (**C**) to (**E**) Depiction of population-level infectiousness curves without (hatched) and with (filled) testing and isolation for the labeled pathogen and testing program, and (**B**) provides annotations for an example simulation. Black curves represent the proportion of infections not yet detected by time *t*. See fig. S7 for scenario ascertainment rates and fig. S8 for population-level infectiousness curves for all testing scenarios.

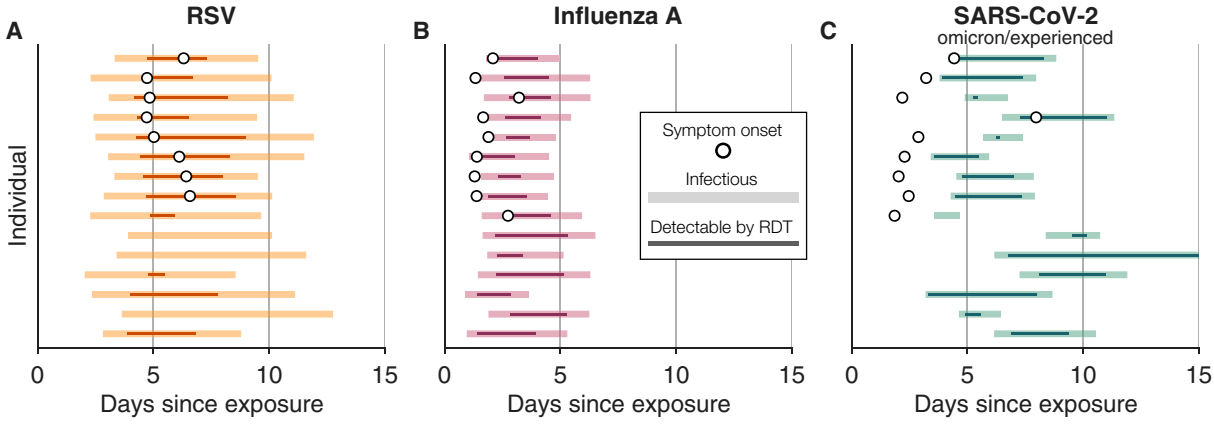

Fig. 3. Relative timing of symptom onset, infectiousness, and detectability vary by pathogen and individual. Symptom onset time (open circles), infectious period (shaded rectangles), and window of detectability by an RDT (colored lines) are shown for 15 stochastic realizations of RSV [(**A**) orange], influenza A [(**B**) pink], and SARS-CoV-2 omicron/experienced [(**C**) green] infections. The absence of an open circle indicates an asymptomatic infection. See table S1 for parameters and references.

their highly overlapping symptom sets, our results indicate that testing immediately after symptom onset with a single three-pathogen RDT (*8*) is therefore likely to mitigate transmission most for influenza A, followed by SARS-CoV-2 omicron and RSV.

This ordering of differential TE was inverted for a single elective RT-qPCR between 2 and 7 days after exposure, which was least effective for influenza A (Fig. 2A). This ordering reflects a substantially faster onset of infectiousness after exposure for influenza A (Fig. 3B), meaning that a postexposure test that is timed effectively for SARS-CoV-2 and RSV is administered far too late to effectively control influenza A. Thus, in a scenario where an individual seeks a highly sensitive multiplex diagnostic test within 1 week after a known exposure to an unknown respiratory pathogen, our results indicate an impact on transmission of 28% for SARS-CoV-2, 22% for RSV, and just 2% for influenza A.

Our results highlight the fact that variation in viral kinetics, symptom onset time, tests' analytical sensitivities, and interactions thereof lead to markedly different impacts on transmission, even under the same testing guidance. Given these complexities, we explored whether varying either the pathogen load above which individuals are considered infectious or the RDT LOD would substantially alter our findings. These sensitivity analyses showed that, while the magnitude of TE varies slightly, the primary trends documented above remain consistent (fig. S3).

Throughout these experiments, we observed that ascertainment—the proportion of infections diagnosed via testing—was only weakly related to TE, showing that a testing program's information value and mitigation impact are distinct quantities. For instance, elective RDT testing after symptoms for influenza A showed TE and ascertainment of 38 and 60%, respectively; for elective RT-qPCR testing after exposure, TE decreased by 36 percentage points to just 2%, but ascertainment decreased by only 10 percentage points to 50% (fig. S1). To illustrate the reason for this difference, we plotted the infectiousness curves $\beta(t)$ with and without testing, averaged over all 10^5 simulated individuals, as well as the curves showing the fraction of individuals remaining undiagnosed $1 - p\ \mathbb{E}[F(t)]$ (Fig. 2, C to E). In instances where diagnoses typically arrive earlier, the average $\beta(t)$ (and thus the area beneath it, R_{testing}) is more substantially reduced, while the same number of diagnoses, arriving later, leave more area under the $\beta(t)$ curve. Thus, TE incorporates not just whether one is diagnosed, but also when.

Impacts of timing and availability of elective post-symptom testing

In an era of increasing elective and self-administered RDT usage, two key questions are when to test and how many tests to use. We sought to answer these questions by modeling the impact of changes in timing and supply of RDTs on TE for RSV, influenza A, and SARS-CoV-2 omicron in experienced hosts. In these experiments, we considered that individuals would wait 0 to 5 days after symptom onset and then begin using one RDT per day, with one to six RDTs available. For comparison, we also computed TE for a single RT-qPCR with a 2-day turnaround. For each fixed supply of tests, we then identified the optimal number of days after symptoms that one should wait to maximize TE separately for each pathogen (Fig. 4, white stars).

This experiment showed three common patterns for RSV and influenza A. First, the most effective timing of post-symptom testing was 0 days, with TE decreasing monotonically for each additional day of delay (Fig. 4, A and B). Second, although using two tests was superior to using one, using more than two tests was roughly equivalent to two. Third, using just one RDT provided superior TE to a single RT-qPCR with a 2-day TAT, highlighting the importance of RDT availability and rapid results for transmission control.

In contrast, for SARS-CoV-2 omicron in experienced hosts, the number of available tests markedly shifted the optimal time at which one should begin testing such that when only one to two RDTs were available, daily testing was most effective beginning 2 days after symptoms; with three to four RDTs, 1 day; and with five to six RDTs, testing should begin immediately upon symptom onset (Fig. 4C). These results reflect a trade-off arising from the variable gap between symptom onset and first detectability by RDTs (Fig. 3C): Using tests later improves the probability of diagnosis but decreases the impact per diagnosis. A large test supply alleviates this trade-off by enabling one to test early, with the potential for high impact, while also testing later to avoid missing a diagnosis entirely. Furthermore, for a fixed delay before testing, using more SARS-CoV-2 RDTs was always superior, yet a single RT-qPCR on the first day of symptoms

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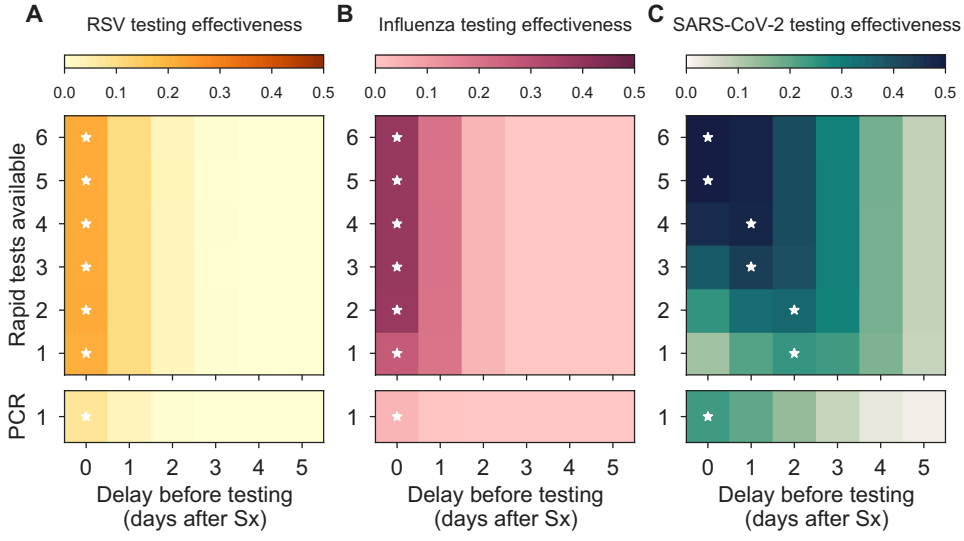

Fig. 4. Optimal use of tests depends on the number of tests available and when they are used. Testing effectiveness (TE) of rapid test (RDT) and RT-qPCR with 2-day TAT, used *x* days after symptom (Sx) onset using *y* tests once per day is shown for RSV [(**A**) orange], influenza A [(**B**) pink], and SARS-CoV-2 omicron in experienced hosts [(**C**) green]. Darker colors represent higher TE as indicated. In each row, the testing strategy with the highest TE is annotated with a white star. TATs: rapid tests, TAT = 0; RT-qPCR, TAT = 2. See table S1 for LODs and figs. S4 and S5 for monochromatic TE and ascertainment visualizations, respectively.

provided approximately equivalent TE to using one RDT starting on day 2.

Together, these results suggest a unified recommendation for the timing of post-symptom testing via RT-qPCR for all three pathogens and post-symptom RDTs for RSV and influenza A: One should test immediately regardless of the number of available tests. In contrast, the timing of optimal SARS-CoV-2 RDT use in supply-limited scenarios depends on the number of available tests. In sensitivity analyses varying the RDT LODs and changing the pathogen load threshold for infectiousness, these general findings were unchanged (fig. S6), although the precise recommended post-symptom wait time for SARS-CoV-2 RDTs shifted in some sensitivity analyses by up to 1 day.

One common point surfaced by our investigations of elective post-symptom testing was that the existence of any asymptomatic and presymptomatic transmission implies that TE < 1 for symptom-driven testing, regardless of the quality of the diagnostic test itself. This implies that even groundbreaking advances in diagnostic LODs, cost, or TATs must be paired with appropriate recommendations for usage.

Last, we briefly note that identical questions of when to test and how many tests to use also arise after known exposure to a pathogen. An otherwise identical analysis recommends wait times of 3 to 5 days after RSV or SARS-CoV-2 exposure and 0 to 2 days after influenza A exposure, depending on the number and type of available tests (fig. S7).

Reevaluation of the sensitivity/turnaround trade-off for SARS-CoV-2

Modeling studies in 2020 and 2021 argued that test sensitivity was secondary to frequency and turnaround time for SARS-CoV-2 screening (*13*, *14*, *47*, *48*) using a similar model which incorporates within-host dynamics and RDT sensitivities for the founder strain in naive hosts. However, three important observations regarding the omicron variants circulating in 2023 led us to revisit these findings. First, studies of viral load trajectories, estimated via prospective longitudinal sampling, show lower peak viral loads, shorter clearance times, and slightly longer proliferation times for omicron infections in experienced hosts compared to founder-strain infections in naive hosts (*17*, *34*), leading to shorter windows of detectability (Fig. 5A). Second, symptom onset is typically 3 to 5 days earlier for omicron/experienced versus founder/naive, relative to peak viral load, an observation argued to be due to the immune experience of hosts (*34*, *49*, *50*). Third, the analytical sensitivity of RDTs is estimated to have worsened for omicron variants versus founder strain, with LODs increasing by 0.5 to 1.0 orders of magnitude depending on the test (*45*, *46*). Together, these factors led us to hypothesize that the previously established superiority of RDTs over RT-qPCR for mitigation of founder-strain SARS-CoV-2 in a naive population could be equalized or reversed for omicron-variant SARS-CoV-2 in an immune experienced population.

To test our hypothesis, we estimated TE for RDT testing (TAT = 0) and RT-qPCR testing (TAT = 2) programs, in twice-weekly, weekly, and elective post-symptom testing scenarios, and compared our findings for founder/naive versus omicron/experienced parameters. We found that in each of the three founder/naive testing scenarios, RDTs provided higher TE than RT-qPCR (Fig. 5B), replicating the claims of the literature (*13*, *14*, *47*, *48*). However, each of the three omicron/experienced scenarios saw a reversal, with RT-qPCR providing higher TE than RDTs, despite the modeled 2-day RT-qPCR TAT (Fig. 5C). In general, we also observed that TE decreased for RDTs from the founder-strain era to the omicron era (Fig. 5, B versus C), while staying approximately the same (twice weekly and weekly) or even increasing (after symptom) for RT-qPCR. Together, these results suggest that, setting aside any differences in cost or regulatory complexity, RT-qPCR–based SARS-CoV-2 testing would be superior to otherwise identical RDT-based testing in the omicron and immune-experienced era.

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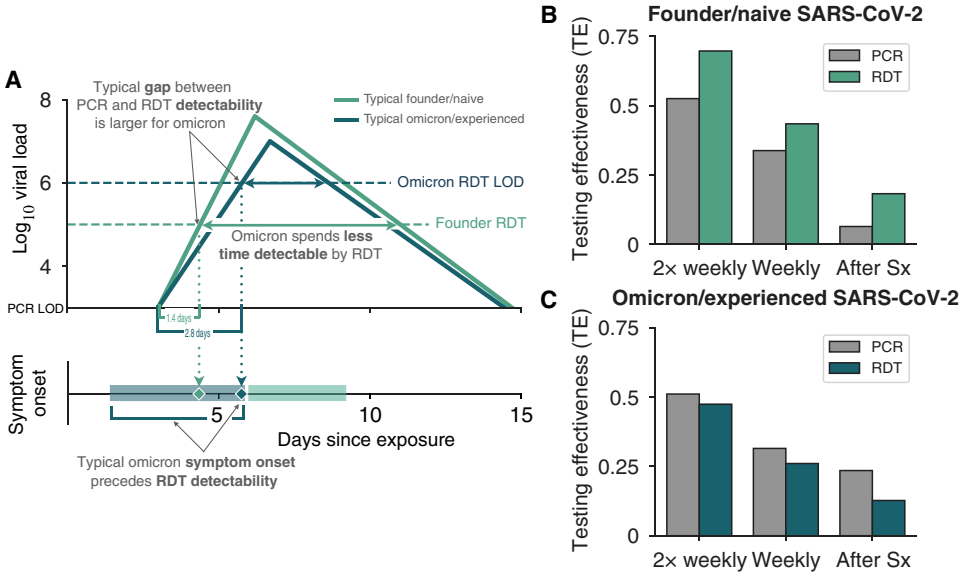

Fig. 5. RT-qPCR versus RDT trade-offs for the SARS-CoV-2 omicron era. (**A**) Typical viral kinetics for founder SARS-CoV-2 strains in naive hosts and SARS-CoV-2 omicron variants in experienced hosts ($\log_{10}$ cp RNA/ml). Trajectories are characterized above the RT-qPCR LOD (= 10^3) with their respective RDT LODs indicated by horizontal dashed lines (10^5 and 10^6 for founder strain and omicron variant, respectively). Individuals are considered infectious when the viral load exceeds $10^{5.5}$ cp RNA/ml. Potential symptom onset times for each trajectory are shaded on the lower axis. (**B** and **C**) TE using RT-qPCR with 2 day TAT (gray) or RDT with immediate delivery of results (green) for twice weekly and weekly screening or testing immediately upon symptom (Sx) onset using one test.

To what can we attribute this apparent reversal in the prioritization of speed versus sensitivity and how might such principles generalize? First, we note that during viral proliferation, there exists a gap between the time of first detectability via RT-qPCR and the time of first detectability via RDT. This gap represents a potential diagnostic advantage for the RT-qPCR test, but it can be realized only if (i) a test is actually taken during the gap, and (ii) the TAT for the RT-qPCR is smaller than the gap. For founder-strain SARS-CoV-2 in the naive host, the typical gap is just 1.4 days, compared to 2.8 days for SARS-CoV-2 omicron variants in experienced hosts (Fig. 5A). Thus, changes in viral proliferation and RDT LODs lead to a longer "head start" for RT-qPCR in the era of omicron variants. Second, we note that shortening the window of analytical detectability reduces the chance of diagnosis. For founder-strain SARS-CoV-2 in the naive host, the typical RDT detection window lasts 6.6 days, versus 2.2 days for SARS-CoV-2 omicron variants in experienced hosts. Thus, RDTs in the omicron variant era have fewer opportunities to contribute to TE than those in the founder strain era. We expect both the head start and detection window intuitions to generalize yet caution that direct estimation of TE is superior to intuition, as not all infections contribute equally to transmission, so an apparent advantage for one test over another in the average or modal infection may disappear or reverse in the most consequential infections.

Estimating costs: Isolation days and test consumption

Diagnosis-driven isolation provides a mitigation benefit of TE but at the cost of testing resources and days spent in isolation. Therefore, in addition to quantifying the benefits of TE and ascertainment, we estimated the costs of test consumption and isolation days (Materials and Methods). When calculating these costs, we also recognized that diagnostic tests may be used to determine when one should exit isolation, via the so-called test-to-exit (TTE) guidelines (*51*), potentially increasing test consumption to decrease isolation days, with the additional risk of early release from isolation due to a nonanalytical test failure. On the basis of these modeling needs, we modified our estimates of all model outputs to incorporate TTE strategies (Supplementary Text).

To explore the ways in which our model could assist in evaluating complex cost-benefit trade-offs across scenarios, we considered elective post-symptom testing for SARS-CoV-2 omicron variants in experienced hosts using either one or two RDTs daily after symptom onset to attempt diagnosis. Diagnosis was followed by either a fixed-duration isolation of 5 days, the minimum time recommended by the US Centers for Disease Control and Prevention at the time of writing (*51*), or a TTE policy of daily testing using the same RDT, beginning after a minimum of 2 days spent in isolation. In each of the 2 × 2 scenarios, we computed TE, the average number of tests consumed per infected individual, and the average number of isolation days per diagnosed individual.

At a high level, using up to two tests to diagnose led to substantially higher TE than using just one (Fig. 6A), mirrored by an increase in per-infection testing cost from 1 to 1.8 tests for the fixed-isolation scenario and from 1.4 to 2.4 tests for the TTE scenario (Fig. 6B). This testing cost is calculated in expectation across all infected individuals, and those who are not detected by the diagnosing test(s) do not use any further tests. Average isolation days per diagnosis were identically 5 days for the two fixed isolation scenarios by definition and reduced to 3.2 to 3.3 days for TTE (Fig. 6C). Thus, as intended, TTE programs decreased average isolation days without a large impact on TE, driven by the fact that the most common outcome of TTE was release after just 48 to 72 hours while nevertheless maintaining long isolations for those remaining detectable for longer periods (Fig. 6D).

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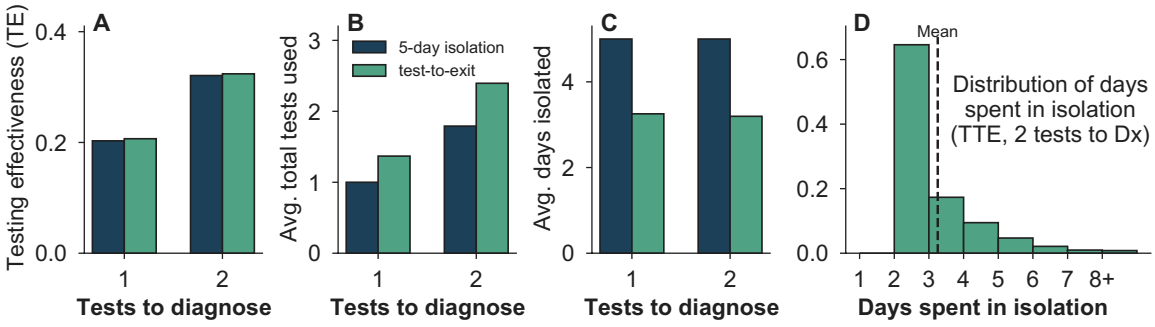

Fig. 6. Fixed isolation recommendations may lead to unnecessary isolation when symptomatic testing ascertainment is high. (**A**) Testing effectiveness, (**B**) total test consumption over diagnosis and test-to-exit (TTE) usage, and (**C**) average isolation duration for detected individuals are shown for a fixed 5-day isolation period (dark blue) and a TTE isolation program requiring one negative RDT to exit isolation (light green) when one or two tests were available to diagnose (Dx), used daily beginning 1 day after symptom onset for SARS-CoV-2 omicron variants in experienced hosts. (**D**) The distribution of individual days spent in isolation for a TTE scheme using two tests to diagnose, with the average isolation time indicated by the vertical dashed line. TTE scenarios assumed individuals waited 2 days after diagnosis before beginning exit testing.

DISCUSSION

This study quantifies the impact of testing as a nonpharmaceutical intervention by defining TE as the expected reduction in the risk of transmission for a particular testing behavior, test, and infecting pathogen. While TE varies considerably across the scenarios we explored, three general principles emerge. First, no single elective testing strategy provides superior control across respiratory viruses due to important differences in their dynamics within the host. This highlights the importance of models linking within-host kinetics to between-host transmission when establishing testing guidelines and the risks of blanket recommendations for respiratory virus transmission control. Second, elective post-symptom or postexposure testing can have a substantial impact on transmission, but timing is important and depends on the pathogen, test, and available supply. While greater availability of tests leads to strictly larger TE, in supply-limited scenarios, a strategic delay may lead to greater TE by diagnosing more infections even if they are not detected as early. Last, by modeling tests and testing through different parameters, our analyses show the importance of key behaviors, including compliance, post-diagnosis isolation, and the manner in which to test. These present opportunities to markedly increase the mitigation impact of testing through not only test technology and availability but through usage guidance and policy as well based on quantitative recommendations from the model.

This work builds on strengths of three established threads in the literature. Many high-quality studies have estimated the value of testing, particularly for routine SARS-CoV-2 screening under a variety of assumptions (*2*, *12*, *14–16*, *18–20*, *47*, *52*) and for HIV (*9*, *22*, *23*, *41*). Recognizing the growing availability of test options for other pathogens (*7*, *8*, *10*) and the collapse of routine SARS-CoV-2 screening programs in favor of elective post-symptom or postexposure testing, our work proposes a more general framework to meet the current and future realities of testing. This work also builds on the concept of symptom-based controllability (*16*, *53*, *54*) by considering a type of diagnosis-driven controllability and quantifying how much of the theoretical value of symptom-based control can be realized when individuals require a diagnosis before deciding to isolate. Third, this work adds TE to the set of estimates quantifying the effectiveness of targeted interventions that can be directly incorporated as single parameters in between-host transmission models, including VE (*55*) and preexposure prophylaxis efficacy (*56*). Conceptually, TE may be implemented as an increased recovery rate (*57*) or decreased transmission probability in between-host models, providing both a computational advantage (TE need be estimated only once for a given testing scenario) and the option to incorporate a variable TE as a parameter in other calculations, for instance, to solve for the TE required to achieve $R_{\text{testing}} < 1$ in a complex compartmental model and then identify testing strategies that achieve it.

By exploring TE under various testing scenarios for three common respiratory viruses, this study demonstrated that an effective strategy for one pathogen may be ineffective for another, and vice versa. This is driven by the complex trade-offs between a test's timing, probability of diagnosis, and number of potentially avoided future transmissions should the test come back positive. Intuitively, early testing leads to fewer diagnoses with higher impact per diagnosis, while delayed testing leads to more diagnoses but lower impact per diagnosis. Mathematical models integrating over these contingent factors, stochasticity, and heterogeneity between individual infections, are critical to putting this intuition on a quantitative foundation.

Exemplifying this value, our analysis showed that the use of RDTs upon symptom onset may be an effective strategy for transmission control of RSV and influenza A, but strategic delays in testing may provide higher effectiveness when test availability is limited for SARS-CoV-2. Our analysis also showed that testing with a slower but more sensitive RT-qPCR exhibited a higher TE than otherwise identical rapid antigen testing for SARS-CoV-2 omicron variants in experienced hosts (but not founder SARS-CoV-2 strains in naive hosts), updating a previously published finding that emphasized TAT over sensitivity for founder-strain SARS-CoV-2 (*13*). This analysis focused only on TE under an assumption of identical participation and compliance. However, self-administered rapid diagnostics may have greater availability, lower cost, and higher tolerability than RT-qPCR tests, factors which may lead to differences in participation and compliance that in turn could be incorporated into updated TE estimates. These examples show how TE and its modeling framework may empower researchers or public health strategists to prospectively evaluate diverse testing strategies and behaviors.

The model introduced here was necessary because empirical evaluation of the impact of a testing program or behavior is difficult, with

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compelling analyses nevertheless lacking formal controls (*2*, *3*) or requiring enormous scale (*1*). In contrast with various methods to empirically estimate *VE*, we know of no direct identification strategies for TE (or similar quantities) in real-world settings. Complicating matters, our calculations suggest that any effective testing program will also shorten the generation interval and increase ascertainment, two quantities that affect the data and models underyling common estimators of the reproductive number *R* (*58*). This suggests that an *R*-based empirical TE estimator would present challenges. Methodological development in this direction would be valuable.

Our modeling provides four potentially useful outputs beyond TE. First, we have attempted to quantify the twin costs of test consumption and isolation days, enabling the Pareto frontier between cost and benefit to be explored under a variety of assumptions. When weighted by dollar costs, these estimates may be useful for individuals and policymakers alike. Second, our calculations estimate ascertainment, which may be useful for surveillance and situational awareness. Third, the distribution of t_{Dx}-TAT, the postexposure time at which the diagnosing test was taken, could be a valuable inclusion in "nowcasting" models, while the implied distribution of pathogen loads in diagnosing samples may be useful in estimating population-scale epidemiologic dynamics (*59*). Last, in smaller populations, actual TE may differ from TE estimates due to finite size effects, variation that could be estimated from sets of Monte Carlo approximations, enabling the incorporation of uncertainty into testing strategy evaluation.

Our work is subject to a number of important limitations associated with the structure of our model. First, we assumed that symptoms may trigger testing but do not trigger isolation in and of themselves. Relaxation of this assumption would require empirical estimates of self-isolation behavior or simply the additional assumption that a proportion of individuals isolate at symptom onset without testing (*13*). Second, we modeled participation/refusal, compliance/adherence, and isolation behaviors as statistically independent between individuals, but health-related behaviors are known to be clustered (*60*–*62*), a type of heterogeneity not included in our TE calculations. In principle, estimates of TE for a set of behaviorally distinct groups could be computed and integrated into appropriately structured transmission models (*57*). Last, our model includes no treatment of specificity, an important factor for certain classes of diagnostic tests. The inclusion of imperfect specificity in our model would not affect TE estimates but could substantially affect cost estimates, particularly if a low positive predictive value led to markedly more isolation days. For such scenarios, our model could be easily extended to include a second confirmatory test to derive a new distribution for t_{Dx}.

Our work is also subject to limitations associated with parameterization. For instance, we relied on estimates of pathogen load dynamics and symptom onset time, including variation thereof across a population. These are relatively well characterized for variants of SARS-CoV-2 in moderate to large population cohorts (*17*, *33*, *35*, *36*, *63*) but are sparse for RSV (*28*, *29*) and influenza A (*30*–*32*) where they come typically from small numbers of healthy volunteers in human challenge studies and are often presented through population means and confidence intervals, which, at best, indirectly inform between-host variation (*64*). Even weakly characterized distributions are effectively unknown for many other pathogens, particularly before symptom onset. Joint estimates of viral kinetics and symptom prevalence and timing for many communicable diseases would be powerful for our modeling, and useful in many other applications as well, including studies of the value of early diagnosis as a path to timely treatment (*20*, *21*). Another important limitation is the assumption that infectiousness can be parameterized as a function of the logarithm of pathogen load. While pathogen load and infectiousness are empirically linked in some studies (*24*, *36*, *65*), their precise relationship is more complicated. For instance, studies of SARS-COV-2 have shown fewer plaque-forming units (PFUs) per copy of viral RNA during an infection's clearance phase than during its proliferation phase (*50*).

Last, as with any public health intervention, there are important ethical considerations related to this work. This study focused on using resources (tests and isolation days) to provide a benefit (decreased transmission), yet not all individuals, communities, or settings may be equally well positioned to afford tests or isolation time. Similarly, our modeling focuses on population-scale transmission but ignores heterogeneity in vulnerability among those to whom a disease may be transmitted. Last, stigma around infection status could lead to low participation, particularly if that otherwise private status will be disclosed (*66*). Application of this work should consider affordability, incentives, vulnerability, and privacy in local contexts.

MATERIALS AND METHODS

Mathematical model for testing

We introduce testing effectiveness (TE) as the proportion by which a testing program decreases the risk of transmission, given infection, for an infectious disease (Eq. 1) and define a mathematical model to prospectively estimate it. This model combines known or assumed properties of (i) a particular diagnostic test, (ii) a strategy for its administration, (iii) behavior/isolation after diagnosis, and (iv) the time-varying profiles of infectiousness and detectability over the course of an infection. Because of its integration of these elements, the model can also estimate a testing program's ascertainment (the proportion of infections detected), the impact of testing on the generation interval and selection coefficients, the distribution of diagnosis times, and the expected number of tests and isolation days required per diagnosis.

Infectiousness

In the absence of testing, the individual reproductive number ν_0 quantifies the expected number of secondary infections from that person (*67*), under typical behavior. It is given by the area under that individual's infectiousness curve $\beta(t)$ over time,

$$\nu_0 = \int_0^\infty \beta(t)\ dt \tag{2}$$

The mean across all individual reproductive numbers is *R* by definition, and therefore

$$R_{\text{no testing}} = \mathbb{E}[\nu_0] = \int_0^\infty \mathbb{E}[\beta(t)]\ dt \tag{3}$$

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By prompting a post-diagnosis behavior change, participation in testing may isolate or attenuate part of that individual's infectiousness, decreasing its total from ν_0 to $\overline{\nu}_{\text{testing}}$. Provided that one's ν_0 and one's testing behaviors are statistically independent, the mean across all individuals' $\overline{\nu}_{\text{testing}}$ is related to the effective reproductive number as

$$R_{\text{testing}} = p\mathbb{E}[\overline{\nu}_{\text{testing}}] + (1-p)\mathbb{E}[\nu_0] \tag{4}$$

where p represents the proportion of the population participating in testing. The participation rate incorporates both the proportion of the population that "opts in" to testing, as well as the fact that even among those opting in, not all will experience symptoms (for elective post-symptom testing) or be alerted to their having been exposed (for elective postexposure testing). Our focus in the derivation that follows is to estimate $\overline{\nu}_{\text{testing}}$ for each person, in order to compute R_{testing} and thus TE.

Isolation after diagnosis

Suppose that at time t_{Dx}, an individual receives a diagnosis that causes them to attenuate their infectiousness via a change in behavior, such as isolation, masking, or increased ventilation. Here, we develop the mathematics corresponding to perfect isolation after diagnosis but provide equations for partial and/or time-varying impacts of behavior on transmission in the Supplementary Materials. Post-diagnosis isolation leads to

$$\nu_{\text{testing}}(t_{\text{Dx}}, t_{\text{exit}}) = \int_0^{t_{\text{Dx}}} \beta(t)\ dt + \int_{t_{\text{exit}}}^{\infty} \beta(t)\ dt \tag{5}$$

which has the simple interpretation that one's infectiousness is that which occurred before diagnosis at t_{Dx} and that which occurred after exiting isolation at t_{exit}. While more complicated test-to-exit equations are developed in the Supplementary Materials, here, we consider only fixed isolation periods of duration ℓ, such that $t_{\text{exit}} = t_{\text{Dx}} + \ell$.

Time of diagnosis as a random variable

In practice, the time of diagnosis t_{Dx} depends on numerous factors including test availability or schedule or the timing of symptoms. We model t_{Dx} as a random variable with probability density function $f(t_{\text{Dx}})$. This leads to an effective infectiousness, calculated in expectation over the probable times of diagnosis, of

$$\overline{\nu}_{\text{testing}} = \int_0^{\infty} \nu_{\text{testing}}(t_{\text{Dx}})\ f(t_{\text{Dx}})\ dt_{\text{Dx}} \tag{6}$$

Substituting in the definition of $\nu_{\text{testing}}(t_{\text{Dx}})$ from eq. S1, rearranging, and making use of the cumulative probability of a diagnosis $F(t)$

$$\overline{\nu}_{\text{testing}} = \nu_0 - \int_0^{\infty} \beta(t)[F(t) - F(t-\ell)]\ dt \tag{7}$$

We note that if one reinterprets $F(t)$ as the cumulative probability of symptom onset (as if symptoms were the diagnostic test itself), with an exhaustive isolation thereafter (large ℓ), then Eq. 7 recovers the core notion of symptom-based controllability of Fraser *et al.* (*53*). However, unlike symptom onset, the time of diagnosis via testing t_{Dx} depends on within-host kinetics, test administration, and of course the test itself, as we calculate next.

Calculating t_{Dx}: Testing, compliance, failure rates, detectability, and TAT

To calculate the distribution of diagnosis times $f(t_{\text{Dx}})$, we combine test administration—the probability that a test is administered at a particular time—and detection—the conditional probability that said test would return a positive diagnosis. This approach naturally separates a test administration strategy from the properties and performance of the particular diagnostic to be modeled.

Let $A(t)$ be the prescribed rate of test administration in tests per day such that for sufficiently small δ

$$\Pr(\text{test administered between } t \text{ and } t+\delta) = \int_t^{t+\delta} A(t')\ dt' \tag{8}$$

For instance, setting $A(t) = \frac{1}{7}$ would model a weekly screening regimen, while $A(t) = 2$ for $4 \le t \le 6$ would represent twice-daily testing starting 4 days after exposure and ending on day 6.

Let $D(t)$ be an indicator function representing the analytical detectability of the infection at time t for a given diagnostic such that $D(t) = 1$ when an infection is, in principle, detectable by that diagnostic, and $D(t) = 0$ otherwise. For example, to model qPCR, $D(t) = 1$ when concentrations of RNA or DNA from a biospecimen exceed the assay's LOD; below the LOD, $D(t) = 0$.

Combining administration with detectability, the cumulative number of scheduled tests with the potential to return a positive result by time t is

$$m(t) = \int_0^{t-\text{TAT}} A(t')\ D(t')\ dt' \tag{9}$$

Note that the integral's upper limit is shifted by TAT, the amount of time between a test's administration and the return of actionable results. Although any particular person may have taken only an integer number of tests, $m(t)$ computes a real-valued expectation over all testing schedules or phases that match the specified rate $A(t)$.

The function $A(t)$ represents an intended test administration strategy (e.g., a policy or guideline), yet in many circumstances, imperfect compliance may result in missed tests. To account for compliance, we let c be the independent Bernoulli probability that each test is actually taken as intended.

Similarly, a test may fail for reasons unrelated to the analytical LOD or presence of a symptom. For example, poor biospecimen collection technique (e.g., poor nasal swab technique) could result in test failure irrespective of an assay's LOD (*45*). To account for test failure of this type, we let ϕ be the independent Bernoulli probability that a test fails for nonanalytical reasons.

Combining the number of possible positive tests taken $m(t)$ with compliance c and test failure ϕ allows us to compute a cumulative distribution function (CDF) representing the probability that one has received a positive test by time t

$$\begin{aligned} F(t) = 1 &- [1-\overline{m}(t)]\ [1-c(1-\phi)]^{m(t)-\overline{m}(t)} \\ &- \overline{m}(t)[1-c(1-\phi)]^{m(t)-\overline{m}(t)+1} \end{aligned} \tag{10}$$

where $\overline{m}(t) = m(t) \bmod 1$. This function reflects the cumulative probability that for a monotonically increasing number of possible positive tests $m(t)$, there are two possibilities: One may have taken

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either $m(t) - \overline{m}(t)$ tests with probability $\left[1 - \overline{m}(t)\right]$ or $m(t) - \overline{m}(t) + 1$ tests with probability $\overline{m}(t)$. In turn, each of these integer number possible positive tests tests may fail (with probability ϕ) or be missed because of noncompliance (with probability *1-c*). We note that this is an improper CDF, which need not reach 1 in the limit of large t, as not all individuals will necessarily receive a diagnosis.

In the case where the product $A(t)D(t)$ is a constant $\overline{A}$ between t_1 and t_2, and 0 otherwise, the associated improper probability density function (PDF) is

$$f(t) = \overline{A}c(1-\phi)\left[1-c(1-\phi)\right]^{m(t)-\overline{m}(t)} \tag{11}$$

for $t_1 + \mathrm{TAT} \leq t \leq t_2 + \mathrm{TAT}$ and 0 otherwise.

In the specific case of elective testing at symptom onset, the functions above require slight modification to include the symptom onset time t_{Sx}. We modeled t_{Sx} as a random offset from the time of peak pathogen load, drawn from a specified uniform distribution (see table S1), a choice that reflects the way symptom onset is typically reported in the literature. For the cases considered here, individuals wait x days after symptoms and test for y days at rate $\overline{A}$ per day, leading to a test administration function

$$A(t, t_{\mathrm{Sx}}) = \overline{A}[u(t - t_{\mathrm{Sx}} - x) - u(t - t_{\mathrm{Sx}} - x - y)] \tag{12}$$

where $u(t)$ is the unit step function defined as 0 before t and 1 thereafter. This equation means that Eq. 9 depends on t_{Sx}, and thus so does the cumulative probability of detection

$$\begin{aligned} F(t, t_{\mathrm{Sx}}) = 1 - \left[1-\overline{m}(t, t_{\mathrm{Sx}})\right][1-c(1-\phi)]^{m(t,t_{\mathrm{Sx}})-\overline{m}(t,t_{\mathrm{Sx}})} \\ -\overline{m}(t, t_{\mathrm{Sx}})[1-c(1-\phi)]^{m(t,t_{\mathrm{Sx}})-\overline{m}(t,t_{\mathrm{Sx}})+1} \end{aligned} \tag{13}$$

The cumulative probability of detection can then be computed by integrating Eq. 13 against the t_{Sx} distribution.

Model estimates

Testing effectiveness

TE can now be computed by substituting Eq. 7 ($\overline{\nu}_{\mathrm{testing}}$) into Eq. 4 ($R_{\mathrm{testing}}$) into Eq. 1 to get

$$TE = p \int_0^\infty \mathbb{E}\{\beta(t)[F(t) - F(t-\ell)]\}dt \Big/ \int_0^\infty \mathbb{E}[\beta(t)]\ dt \tag{14}$$

where Eq. 10 can be used to compute the CDF $F(t)$ of diagnosis times t_{Dx}. If isolation is of sufficient duration to prevent all post-diagnosis transmission, TE simplifies to

$$TE = p \int_0^\infty \mathbb{E}[\beta(t)F(t)]dt \Big/ \int_0^\infty \mathbb{E}[\beta(t)]\ dt \tag{15}$$

Note also that the integrands enclosed in expectations above have useful interpretations: They represent the expected infectiousness trajectories with (numerator) and without (denominator) participation in testing.

Ascertainment, and diagnosis, and swab time distributions

For a single individual, the distribution of diagnosis times is given by its improper PDF $f(t_{\mathrm{Dx}})$ or CDF $F(t)$. The related improper distribution of "diagnosing swab" times is given by shifting the CDF by TAT.

The long-time limit of $F(t)$ is the probability that that individual will be diagnosed at all. Ascertainment, the proportion of infections diagnosed by a particular strategy at the population scale, can therefore be estimated by taking the expectation of the long-time limits over individuals and scaling by the participation rate,

$$\alpha = p \lim_{t \to \infty} \mathbb{E}[F(t)] \tag{16}$$

Here, the expectation is taken over individual heterogeneity in within-host dynamics across the population (just as in Eqs. 3 and 4).

Mean generation interval and selection coefficient

The mean generation interval, defined as the typical time between infection and subsequent transmission, can be computed from the centers of mass of the expected infectiousness curves that appear in the numerator and denominator of Eq. 14. Such estimates may be important for post hoc estimation of TE from empirical data, given that methods of estimating R from empirical case counts often rely on the mean generation interval.

If a single test is capable of diagnosing two strains of the same pathogen, then there is the potential for testing to alter the selection coefficient. Here, TE calculations can estimate this impact such that the selection coefficient is predicted to shift from $s = \frac{R_A}{R_B}$ to $s_{\mathrm{testing}} = s\frac{(1-TE_A)}{(1-TE_B)}$.

Test consumption

How many tests are consumed during the course of a testing regimen? We break this calculation into two cases depending on whether the individual receives a diagnosis or not.

First, suppose an individual never tests positive. This implies that any tests that were meant to be taken when the infection was detectable, i.e., when $D(t) = 1$, were either (i) not taken due to a failure of compliance (probability *1-c*) or (ii) taken yet failed to provide the correct positive diagnosis (probability $c\phi$). Because only the latter case consumes a test, this means that a key quantity is $c\phi/(1 - c + c\phi)$, the relative probability that a planned test consumed a test, given the absence of diagnosis. On the other hand, any tests meant to be taken when the infection was undetectable, i.e., when $D(t) = 0$, were taken with probability c. Summing these two provides an estimate of average test consumption for those who are not diagnosed

$$\overline{q}_{\mathrm{no-Dx}} = \frac{c\phi}{1 - c + c\phi} \int_0^\infty A(t)D(t)\ dt + c\int_0^\infty A(t)[1 - D(t)]\ dt \tag{17}$$

The reason that we call this an estimate and not an exact calculation is that testing and diagnosis are causally linked, and therefore conditioning on a no-diagnosis outcome means that the prescribed test administration may no longer be $A(t)$ exactly.

In contrast, if an individual receives a diagnosis at t_{Dx}, we assume that no additional tests were consumed thereafter and that at least one test was consumed—if not, no diagnosis could have been produced. For a fixed time of diagnosis,

$$q_{\mathrm{Dx}}(t_{\mathrm{Dx}}) = 1 + c\left[\left[\int_0^{t_{\mathrm{Dx}}-\mathrm{TAT}} A(t)\ dt\right] + \left[\int_{t_{\mathrm{Dx}}-\mathrm{TAT}}^{t_{\mathrm{Dx}}} A(t)\ dt\right]\right] \tag{18}$$

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The first integral accounts for nondiagnostic test consumption up until the diagnosing test, while the second integral accounts for additional tests consumed while waiting for the diagnosing test. The separate floor functions are a necessary consequence of conditioning on the separate counting of the diagnosing test. Taking an expectation over t_{Dx} yields

$$\overline{q}_{\mathrm{Dx}} = 1 + \frac{pc}{\alpha}\left[\int_0^{\infty}\left[\int_0^{t_{\mathrm{Dx}}-\mathrm{TAT}} A(t)\ dt\right] f(t_{\mathrm{Dx}})\ dt_{\mathrm{Dx}} + \int_0^{\infty}\left[\int_{t_{\mathrm{Dx}}-\mathrm{TAT}}^{t_{\mathrm{Dx}}} A(t)\ dt\right] f(t_{\mathrm{Dx}})\ dt_{\mathrm{Dx}}\right] \tag{19}$$

Combining our calculations for $\overline{q}_{\mathrm{no-Dx}}$ and $\overline{q}_{\mathrm{Dx}}$, weighting the latter by ascertainment and the former by its complement, we get a general expression for the expected test consumption among infected individuals

$$\overline{q}_{\mathrm{infected}} = (1-\alpha)\ \overline{q}_{\mathrm{no-Dx}} + \alpha\ \overline{q}_{\mathrm{Dx}} \tag{20}$$

Of course, the whole point of testing is that one does not know a priori who is infected and who is not. Consequently, we can estimate consumption by those who are not infected as

$$\overline{q}_{\mathrm{notinfected}} = c\int_0^{\infty} A(t)dt \tag{21}$$

Last, combining Eqs. 20 and 21, we get

$$Q = p[(\theta\ \mathrm{se}_{\mathrm{ref.}})\ \overline{q}_{\mathrm{infected}} + (1-\theta)(1-\mathrm{sp}_{\mathrm{ref.}})\ \overline{q}_{\mathrm{not\,infected}}] \tag{22}$$

where p is the population participation rate, θ is the prevalence of the pathogen, and $\mathrm{se}_{\mathrm{ref.}}$ and $\mathrm{sp}_{\mathrm{ref.}}$ are the sensitivity and specificity of the scheme used to refer people to testing, respectively. Such referral schemes include contact tracing, the appearance of symptoms, membership in a group with known high risk of infection, or even universal testing (in which case, the referral "program" would have se = 1 and sp = 0).

Days spent in isolation

For fixed-duration perfect isolation of length ℓ, the average number of isolation days per infected person is given by the product of ℓ and ascertainment $d = \ell\alpha$. A similar calculation for TTE policies is found in eq. S10.

Parameterization of the model

Viral kinetics, symptoms onset, and infectiousness

We parameterized the time-varying profiles of viral load (VL) and infectiousness using a simplified model that captures four features common to RSV (*28*), influenza A (*30*), and multiple variants of SARS-CoV-2 (*17*): (i) a postexposure period where virus is undetectable by any known test, (ii) a proliferation period of exponential growth, (iii) a peak VL, followed by (iv) a clearance period of exponential decline. We capture these features using a piecewise linear function describing the logarithm of VL, specified by three points: (t_0,LLOQ), the first time at which VL exceeds the lower limit of quantification (LLOQ); (t_{peak},V_{peak}), peak VL timing and concentration; and (t_f,LLOQ), the last detectable time (*13*, *17*), where t represents the time since exposure in days (fig. S8). For the pathogens considered here, the LLOQ is equal to the RT-qPCR LOD. Because within-host kinetics vary from one infection to another, each trajectory is parameterized using independent draws from random variables for each control point.

While summarized in table S1, we briefly review the studies and sources of evidence used to parameterize our simple VL models. Influenza A latent period parameters were drawn from a review of challenge studies by Carrat *et al.* (*30*) supporting a 0.5- to 2-day delay between inoculation and first detectable instance using a gold standard test, noting that this range is slightly wider than in other studies (*31*). Data from a household study from Ip *et al.* characterized influenza A peak VL between 1 and 3 days after symptom onset and noted that symptom onset and first detectability were indistinguishable. Their data also showed peak VL between 6 and 8.5 $\log_{10}$ cp RNA/ml, followed by 2- to 3-day clearance times (*32*). We note that these clearance times were shorter than those observed in challenge studies (*30*, *31*). Symptom onset time was specified as taking place between 2 and 0 days before peak VL from observations of naturally acquired infection (*38*).

RSV viral kinetics were characterized using vaccine efficacy challenge studies, which reported only geometric means of VL, measured in days since challenge inoculation (*28*, *29*). Placebo group data from both studies support a 2- to 4-day latent period and 3 to 6 days between peak VL and clearance (*28*, *29*). Schmoele-Thoma *et al.* (*28*) present individual data points for longitudinal sampling of infected participants supporting a proliferation phase of 2 to 4 days and peak VL between 4.5 and 8 $\log_{10}$ cp RNA/ml. Sadoff *et al.* (*29*) indicate slightly later and lower peaks, but we weight these less heavily in model parameterization because only mean and confidence intervals are presented, not individual data points. Symptom onset time was specified as taking place between 1 day before and 1 day after peak VL from an additional human challenge study (*40*).

We briefly note that our characterization of influenza A and RSV viral kinetics relies primarily on studies meant to capture vaccine (*28*, *29*) and drug (*31*) efficacy or to report symptom dynamics (*32*). In all four studies, viral kinetics data are reported as a secondary finding and typically as a geometric mean since time of onset—including, in two instances, all the individuals for whom inoculation failed (*28*, *29*). Because of the lack of VL data linked at the level of individuals to inform distributional choices of kinetics parameters across a population, we assumed uniform distributions over supported parameter ranges for both viruses.

In contrast, our estimates of SARS-CoV-2 viral kinetics parameters were drawn from studies performed specifically to characterize within-host viral dynamics. Kissler *et al.* and Hay *et al.* (*17*, *34*) provide mean and 95% credible interval estimates for peak VL and the durations of the proliferation and clearance phases for founder-strain/naive and omicron-strain/experienced SARS-CoV-2, respectively. Parameter distributions were assumed to be Lognormal with parameters μ and σ, whose values were adapted from published means, 95% credible intervals, and sample sizes. Specifically, we selected values such that an equal number of draws from Lognormal (μ, σ) would lead to a frequentist confidence interval matching the published mean and credible interval to the closest approximation. See table S1 for these parameter values. Given the possibility of the occasional nonbiologically large draw from the Lognormal distributions, we rejected proliferation phases shorter than 0.5 days and longer than 10 days, and rejected clearance phases shorter than 0.5 days and longer than 25 days for both SARS-CoV-2 models. A challenge study provides support for a 2.5- to 3.5-day latent period after inoculation (*33*). Symptom onset time was specified as taking place between 0 and 3 days after peak VL for founder-strain/naive infections (*34*, *49*, *50*), and between 1 and 5 days before peak VL for omicron-variant/experienced infections (*33*, *68*, *69*).

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Given a stochastic realization of viral kinetics from the model above, we calculated infectiousness $\beta(t)$ as proportional to the logarithm of VL in excess of some minimum threshold, specified by the typical VL (concentration of RNA, cpRNA/ml) at which PFUs are consistently found (>1 PFU/ml). While this type of log-viral-load infectiousness assumption is common for studies of influenza A (*30*, *70*), SARS-CoV-2 (*52*), and RSV (*71*), alternative relationships between VL (or other quantities) and infectiousness are possible (*13*, *72*).

Analytical sensitivity and failure rate of diagnostic tests

Analytical sensitivities (LODs) were drawn from the literature for RT-qPCR and RDT tests for influenza A (*73*), RSV (*74*), founder-strain SARS-CoV-2, and SARS-CoV-2 omicron variants (*75*–*77*). Because of the variability in LODs between assays of the same type for the same pathogen, we chose a single value (table S1) to represent each pathogen and test type. Above a test's estimated LOD, false negative rates ϕ were available only for SARS-CoV-2 (*45*) at approximately 5%, a rate we assumed for RSV and influenza A, collectively modeling factors such as sample contamination, poor biospecimen collection, or manufacturing error. In general, parameters were more widely available and better characterized for SARS-CoV-2 tests than for RSV or influenza A. See table S1 for LODs, failure rates, and relevant sources.

Supplementary Materials

This PDF file includes:

Supplementary Text
Figs. S1 to S8
Table S1
Equations S1 to S10

REFERENCES AND NOTES

1. B. E. McGarry, A. D. Gandhi, M. L. Barnett, COVID-19 surveillance testing and resident outcomes in nursing homes. *N. Engl. J. Med.* **388**, 1101–1110 (2023).
2. D. R. E. Ranoa, R. L. Holland, F. G. Alnaji, K. J. Green, L. Wang, R. L. Fredrickson, T. Wang, G. N. Wong, J. Uelmen, S. Maslov, Z. J. Weiner, A. V. Tkachenko, H. Zhang, Z. Liu, A. Ibrahim, S. J. Patel, J. M. Paul, N. P. Vance, J. G. Gulick, S. P. Satheesan, I. J. Galvan, A. Miller, J. Grohens, T. J. Nelson, M. P. Stevens, P. M. Hennessy, R. C. Parker Jr., E. Santos, C. Brackett, J. D. Steinman, M. R. Fenner Jr., K. Dohrer, M. De Lorenzo, L. Wilhelm-Barr, B. R. Brauer, C. Best-Popescu, G. Durack, N. Wetter, D. M. Kranz, J. Breitbarth, C. Simpson, J. A. Pryde, R. N. Kaler, C. Harris, A. C. Vance, J. L. Silotto, M. Johnson, E. A. Valera, P. K. Anton, L. Mwilambwe, S. P. Bryan, D. S. Stone, D. B. Young, W. E. Ward, J. Lantz, J. A. Vozenilek, R. Bashir, J. S. Moore, M. Garg, J. C. Cooper, G. Snyder, M. H. Lore, D. L. Yocum, N. J. Cohen, J. E. Novakofski, M. J. Loots, R. L. Ballard, M. Band, K. M. Banks, J. D. Barnes, I. Bentea, J. Black, J. Busch, A. Conte, M. Conte, M. Curry, J. Eardley, A. Edwards, T. Eggett, J. Fleurimont, D. Foster, B. W. Fouke, N. Gallagher, N. Gastala, S. A. Genung, D. Glueck, B. Gray, A. Greta, R. M. Healy, A. Hetrick, A. A. Holterman, N. Ismail, I. Jasenof, P. Kelly, A. Kielbasa, T. Kiesel, L. M. Kindle, R. L. Lipking, Y. C. Manabe, J. Mayes, R. M. Guffin, K. G. M. Henry, A. Mirza, J. Moseley, H. H. Mostafa, M. Mumford, K. Munoz, A. D. Murray, M. Nolan, N. A. Parikh, A. Pekosz, J. Pflugmacher, J. M. Phillips, C. Pitts, M. C. Potter, J. Quisenberry, J. Rear, M. L. Robinson, E. Rosillo, L. N. Rye, M. E. Sherwood, A. Simon, J. M. Singson, C. Skadden, T. H. Skelton, C. Smith, M. Stech, R. Thomas, M. A. Tomaszewski, E. A. Tyburski, S. Vanwingerden, E. Vlach, R. S. Watkins, K. Watson, K. C. White, T. L. Killeen, R. J. Jones, A. C. Cangellaris, S. A. Martinis, A. Vaid, C. B. Brooke, J. T. Walsh, A. Elbanna, W. C. Sullivan, R. L. Smith, N. Goldenfeld, T. M. Fan, P. J. Hergenrother, M. D. Burke, Mitigation of SARS-CoV-2 transmission at a large public university. *Nat. Commun.* **13**, 3207 (2022).
3. M. Pavelka, K. Van-Zandvoort, S. Abbott, K. Sherratt, CMMID COVID-19 working group, Inštitút Zdravotných Analýz, P. Jarčuška, M. Krajčí, S. Flasche, S. Funk, The impact of population-wide rapid antigen testing on SARS-CoV-2 prevalence in Slovakia. *Science* **372**, 635–641 (2021).
4. D. I. Bernstein, A. Mejias, B. Rath, C. W. Woods, J. P. Deeter, Summarizing study characteristics and diagnostic performance of commercially available tests for respiratory syncytial virus: A scoping literature review in the COVID-19 era. *J. Appl. Lab. Med.* **8**, 353–371 (2023).
5. J. Merckx, R. Wali, I. Schiller, C. Caya, G. C. Gore, C. Chartrand, N. Dendukuri, J. Papenburg, Diagnostic accuracy of novel and traditional rapid tests for influenza infection compared with reverse transcriptase polymerase chain reaction: A systematic review and meta-analysis. *Ann. Intern. Med.* **167**, 394–409 (2017).
6. D. P. Ghumra, N. Shetty, K. R. McBrearty, J. V. Puthussery, B. J. Sumlin, W. D. Gardiner, B. M. Doherty, J. P. Magrecki, D. L. Brody, T. J. Esparza, J. A. O'Halloran, R. M. Presti, T. L. Bricker, A. C. M. Boon, C. M. Yuede, J. R. Cirrito, R. K. Chakrabarty, Rapid direct detection of SARS-CoV-2 aerosols in exhaled breath at the point of care. *ACS Sens.* **8**, 3023–3031 (2023).
7. H.-Y. Chung, M.-J. Jian, C.-K. Chang, J.-C. Lin, K.-M. Yeh, C.-W. Chen, S.-K. Chiu, Y.-H. Wang, S.-J. Liao, S.-Y. Li, S.-S. Hsieh, S.-H. Tsai, C.-L. Perng, J.-R. Yang, M.-T. Liu, F.-Y. Chang, H.-S. Shang, Novel dual multiplex real-time RT-PCR assays for the rapid detection of SARS-CoV-2, influenza A/B, and respiratory syncytial virus using the BD MAX open system. *Emerg. Microbes Infect.* **10**, 161–166 (2021).
8. S. Ferrani, T. Prazuck, S. Béchet, F. Lesne, R. Cohen, C. Levy, Diagnostic accuracy of a rapid antigen triple test (SARS-CoV-2, respiratory syncytial virus, and influenza) using anterior nasal swabs versus multiplex RT-PCR in children in an emergency department. *Infect. Dis. Now* **53**, 104769 (2023).
9. J. S. Haukoos, E. Hopkins, A. A. Conroy, M. Silverman, R. L. Byyny, S. Eisert, M. W. Thrun, M. L. Wilson, A. B. Hutchinson, J. Forsyth, S. C. Johnson, J. D. Heffelfinger, Denver Emergency Department HIV Opt-Out Study Group, Routine opt-out rapid HIV screening and detection of HIV infection in emergency department patients. *JAMA* **304**, 284–292 (2010).
10. D. W. Brown, L. Warrener, H. M. Scobie, M. Donadel, D. Waku-Kouomou, M. N. Mulders, P. A. Rota, Rapid diagnostic tests to address challenges for global measles surveillance. *Curr. Opin. Virol.* **41**, 77–84 (2020).
11. S. M. Feleke, E. N. Reichert, H. Mohammed, B. G. Brhane, K. Mekete, H. Mamo, B. Petros, H. Solomon, E. Abate, C. Hennelly, M. Denton, C. Keeler, N. J. Hathaway, J. J. Juliano, J. A. Bailey, E. Rogier, J. Cunningham, O. Aydemir, J. B. Parr, *Plasmodium falciparum* is evolving to escape malaria rapid diagnostic tests in Ethiopia. *Nat. Microbiol.* **6**, 1289–1299 (2021).
12. A. David Paltiel, A. Zheng, R. P. Walensky, Assessment of SARS-CoV-2 screening strategies to permit the safe reopening of college campuses in the United States. *JAMA Netw. Open* **3**, e2016818 (2020).
13. D. B. Larremore, B. Wilder, E. Lester, S. Shehata, J. M. Burke, J. A. Hay, M. Tambe, M. J. Mina, R. Parker, Test sensitivity is secondary to frequency and turnaround time for COVID-19 screening. *Sci. Adv.* **7**, eabd5393 (2021).
14. T. Bergstrom, C. T. Bergstrom, H. Li, Frequency and accuracy of proactive testing for COVID-19. medRxiv 2020.09.05.20188839 [Preprint] (2020). https://doi.org/10.1101/2020.09.05.20188839.
15. N. C. Grassly, M. Pons-Salort, E. P. K. Parker, P. J. White, N. M. Ferguson, Imperial College COVID- Response Team, Comparison of molecular testing strategies for COVID-19 control: A mathematical modelling study. *Lancet Infect. Dis.* **20**, 1381–1389 (2020).
16. A. J. Kucharski, P. Klepac, A. J. K. Conlan, S. M. Kissler, M. L. Tang, H. Fry, J. R. Gog, W. J. Edmunds, CMMID COVID-19 working group, Effectiveness of isolation, testing, contact tracing, and physical distancing on reducing transmission of SARS-CoV-2 in different settings: A mathematical modelling study. *Lancet Infect. Dis.* **20**, 1151–1160 (2020).
17. S. M. Kissler, J. R. Fauver, C. Mack, S. W. Olesen, C. Tai, K. Y. Shiue, C. C. Kalinich, S. Jednak, I. M. Ott, C. B. F. Vogels, J. Wohlgemuth, J. Weisberger, J. D. Fiori, D. J. Anderson, J. Mancell, D. D. Ho, N. D. Grubaugh, Y. H. Grad, Viral dynamics of acute SARS-CoV-2 infection and applications to diagnostic and public health strategies. *PLoS Biol.* **19**, e3001333 (2021).
18. I. Holmdahl, R. Kahn, J. A. Hay, C. O. Buckee, M. J. Mina, Estimation of transmission of COVID-19 in simulated nursing homes with frequent testing and immunity-based staffing. *JAMA Netw. Open* **4**, e2110071 (2021).
19. R. L. Smith, L. L. Gibson, P. P. Martinez, R. Ke, A. Mirza, M. Conte, N. Gallagher, A. Conte, L. Wang, R. Fredrickson, D. C. Edmonson, M. E. Baughman, K. K. Chiu, H. Choi, T. W. Jensen, K. R. Scardina, S. Bradley, S. L. Gloss, C. Reinhart, J. Yedetore, A. N. Owens, J. Broach, B. Barton, P. Lazar, D. Henness, T. Young, A. Dunnett, M. L. Robinson, H. H. Mostafa, A. Pekosz, Y. C. Manabe, W. J. Heetderks, D. D. McManus, C. B. Brooke, Longitudinal assessment of diagnostic test performance over the course of acute sars-cov-2 infection. *J. Infect. Dis.* **224**, 976–982 (2021).
20. T. F. Menkir, C. A. Donnelly, The impact of repeated rapid test strategies on the effectiveness of at-home antiviral treatments for SARS-CoV-2. *Nat. Commun.* **13**, 5283 (2022).
21. S. Zhang, A. A. Agyeman, C. Hadjichrysanthou, J. F. Standing, SARS-CoV-2 viral dynamic modelling to inform model selection and timing and efficacy of antiviral therapy. *CPT Pharmacometrics Syst. Pharmacol.* **12**, 1450–1460 (2023).
22. J. M. Hyman, J. Li, E. A. Stanley, Modeling the impact of random screening and contact tracing in reducing the spread of HIV. *Math. Biosci.* **181**, 17–54 (2003).
23. Y. Yazdanpanah, C. E. Sloan, C. Charlois-Ou, S. L. Vu, C. Semaille, D. Costagliola, J. Pillonel, A.-I. Poullié, O. Scemama, S. Deuffic-Burban, E. Losina, R. P. Walensky, K. A. Freedberg, A. D. Paltiel, Routine HIV screening in France: Clinical impact and cost-effectiveness. *PLOS ONE* **5**, e13132 (2010).

Downloaded from https://www.science.org on March 08, 2026

24. K. K. Bjorkman, T. K. Saldi, E. Lasda, L. C. Bauer, J. Kovarik, P. K. Gonzales, M. R. Fink, K. L. Tat, C. R. Hager, J. C. Davis, C. D. Ozeroff, G. R. Brisson, D. B. Larremore, L. A. Leinwand, M. B. McQueen, R. Parker, Higher viral load drives infrequent severe acute respiratory syndrome coronavirus 2 transmission between asymptomatic residence hall roommates. *J. Infect. Dis.* **224**, 1316–1324 (2021).
25. L. E. Smith, R. Amlôt, H. Lambert, I. Oliver, C. Robin, L. Yardley, G. J. Rubin, Factors associated with adherence to self-isolation and lockdown measures in the UK: A crosssectional survey. *Public Health* **187**, 41–52 (2020).
26. T. Braeye, L. Cornelissen, L. Catteau, F. Haarhuis, K. Proesmans, K. De Ridder, A. Djiena, R. Mahieu, F. De Leeuw, A. Dreuw, N. Hammami, S. Quoilin, H. Van Oyen, C. Wyndham-Thomas, D. Van Cauteren, Vaccine effectiveness against infection and onwards transmission of COVID-19: Analysis of Belgian contact tracing data, January-June 2021. *Vaccine* **39**, 5456–5460 (2021).
27. D. W. Eyre, D. Taylor, M. Purver, D. Chapman, T. Fowler, K. B. Pouwels, A. S. Walker, T. E. A. Peto, Effect of Covid-19 vaccination on transmission of alpha and delta variants. *N. Engl J. Med.* **386**, 744–756 (2022).
28. B. Schmoele-Thoma, A. M. Zareba, Q. Jiang, M. S. Maddur, R. Danaf, A. Mann, K. Eze, J. Fok-Seang, G. Kabir, A. Catchpole, D. A. Scott, A. C. Gurtman, K. U. Jansen, W. C. Gruber, P. R. Dormitzer, K. A. Swanson, Vaccine efficacy in adults in a respiratory syncytial virus challenge study. *N. Engl. J. Med.* **386**, 2377–2386 (2022).
29. J. Sadoff, E. De Paepe, J. De Vincenzo, E. Gymnopoulou, J. Menten, B. Murray, A. R. Bastian, A. Vandebosch, W. Haazen, N. Noulin, C. Comeaux, E. Heijnen, K. Eze, A. Gilbert, R. Lambkin-Williams, H. Schuitemaker, B. Callendret, Prevention of respiratory syncytial virus infection in healthy adults by a single immunization of Ad26.RSV.preF in a human challenge study. *J. Infect. Dis.* **226**, 396–406 (2022).
30. F. Carrat, E. Vergu, N. M. Ferguson, M. Lemaitre, S. Cauchemez, S. Leach, A.-J. Valleron, Time lines of infection and disease in human influenza: A review of volunteer challenge studies. *Am. J. Epidemiol.* **167**, 775–785 (2008).
31. S. E. Sloan, K. J. Szretter, B. Sundaresh, K. M. Narayan, P. F. Smith, D. Skurnik, S. Bedard, J. M. Trevejo, D. Oldach, Z. Shriver, Clinical and virological responses to a broad-spectrum human monoclonal antibody in an influenza virus challenge study. *Antiviral Res.* **184**, 104763 (2020).
32. D. K. M. Ip, L. L. H. Lau, K.-H. Chan, V. J. Fang, G. M. Leung, M. J. S. Peiris, B. J. Cowling, The dynamic relationship between clinical symptomatology and viral shedding in naturally acquired seasonal and pandemic influenza virus infections. *Clin. Infect. Dis.* **62**, 431–437 (2016).
33. B. Killingley, A. J. Mann, M. Kalinova, A. Boyers, N. Goonawardane, J. Zhou, K. Lindsell, S. S. Hare, J. Brown, R. Frise, E. Smith, C. Hopkins, N. Noulin, B. Löndt, T. Wilkinson, S. Harden, H. M. Shane, M. Baillet, A. Gilbert, M. Jacobs, C. Charman, P. Mande, J. S. Nguyen-van-Tam, M. G. Semple, R. C. Read, N. M. Ferguson, P. J. Openshaw, G. Rapeport, W. S. Barclay, A. P. Catchpole, C. Chiu, Safety, tolerability and viral kinetics during SARS-CoV-2 human challenge in young adults. *Nat. Med.* **28**, 1031–1041 (2022).
34. J. A. Hay, S. M. Kissler, J. R. Fauver, C. Mack, C. G. Tai, R. M. Samant, S. Connolly, D. J. Anderson, G. Khullar, M. M. Kay, M. Patel, S. Kelly, A. Manhertz, I. Eiter, D. Salgado, T. Baker, B. Howard, J. T. Dudley, C. E. Mason, M. Nair, Y. Huang, J. D. Fiori, D. D. Ho, N. D. Grubaugh, Y. H. Grad, Quantifying the impact of immune history and variant on SARS-CoV-2 viral kinetics and infection rebound: A retrospective cohort study. *eLife* **11**, e81849 (2022).
35. R. Ke, C. Zitzmann, D. D. Ho, R. M. Ribeiro, A. S. Perelson, In vivo kinetics of SARS-CoV-2 infection and its relationship with a person's infectiousness. *Proc. Natl. Acad. Sci. U.S.A.* **118**, e2111477118 (2021).
36. R. Ke, P. P. Martinez, R. L. Smith, L. L. Gibson, A. Mirza, M. Conte, N. Gallagher, C. H. Luo, J. Jarrett, R. Zhou, A. Conte, T. Liu, M. Farjo, K. K. O. Walden, G. Rendon, C. J. Fields, L. Wang, R. Fredrickson, D. C. Edmonson, M. E. Baughman, K. K. Chiu, H. Choi, K. R. Scardina, S. Bradley, S. L. Gloss, C. Reinhart, J. Yedetore, J. Quicksall, A. N. Owens, J. Broach, B. Barton, P. Lazar, W. J. Heetderks, M. L. Robinson, H. H. Mostafa, Y. C. Manabe, A. Pekosz, D. D. Mc Manus, C. B. Brooke, Daily longitudinal sampling of SARS-CoV-2 infection reveals substantial heterogeneity in infectiousness. *Nat. Microbiol.* **7**, 640–652 (2022).
37. J. S. M. Peiris, C. M. Chu, V. C. C. Cheng, K. S. Chan, I. F. N. Hung, L. L. M. Poon, K. I. Law, B. S. F. Tang, T. Y. W. Hon, C. S. Chan, K. H. Chan, J. S. C. Ng, B. J. Zheng, W. L. Ng, R. W. M. Lai, Y. Guan, K. Y. Yuen, HKU/UCH SARS Study Group, Clinical progression and viral load in a community outbreak of coronavirus-associated SARS pneumonia: A prospective study. *Lancet* **361**, 1767–1772 (2003).
38. L. L. H. Lau, B. J. Cowling, V. J. Fang, K.-H. Chan, E. H. Y. Lau, M. Lipsitch, C. K. Y. Cheng, P. M. Houck, T. M. Uyeki, J. S. M. Peiris, G. M. Leung, Viral shedding and clinical illness in naturally acquired influenza virus infections. *J. Infect. Dis.* **201**, 1509–1516 (2010).
39. P. K. Munywoki, D. C. Koech, C. N. Agoti, A. Bett, P. A. Cane, G. F. Medley, D. J. Nokes, Frequent asymptomatic respiratory syncytial virus infections during an epidemic in a rural Kenyan household cohort. *J. Infect. Dis.* **212**, 1711–1718 (2015).
40. F. Eun-Hyung Lee, E. E. Walsh, A. R. Falsey, R. F. Betts, J. J. Treanor, Experimental infection of humans with A2 respiratory syncytial virus1. *Antiviral Res.* **63**, 191–196 (2004).
41. Cost-effectiveness of preventing AIDS complications (CEPAC) model. https://massgeneral.org/medicine/mpec/research/cpac-model.
42. World Health Organization. Statement on the fifteenth meeting of the IHR (2005) emergency committee on the COVID-19 pandemic. https://who.int/news/item/05-05-2023-statement-on-the-fifteenth-meeting-of-the-international-health-regulations-(2005)-emergency-committee-regarding-the-coronavirus-disease-(COVID-19)-pandemic.
43. U.S. Centers for Disease Control and Prevention. Preliminary estimated influenza illnesses, medical visits, hospitalizations, and deaths in the United States–2021-2022 influenza season. https://cdc.gov/flu/about/burden/2021-2022.htm.
44. Y. Li, X. Wang, D. M. Blau, M. T. Caballero, D. R. Feikin, C. J. Gill, S. A. Madhi, S. B. Omer, E. A. F. Simões, H. Campbell, A. B. Pariente, D. Bardach, Q. Bassat, J.-S. Casalegno, G. Chakhunashvili, N. Crawford, D. Danilenko, L. A. H. Do, M. Echavarria, A. Gentile, A. Gordon, T. Heikkinen, Q. S. Huang, S. Jullien, A. Krishnan, E. L. Lopez, J. Markić, A. Mira-Iglesias, H. C. Moore, J. Moyes, L. Mwananyanda, D. J. Nokes, F. Noordeen, E. Obodai, N. Palani, C. Romero, V. Salimi, A. Satav, E. Seo, Z. Shchomak, R. Singleton, K. Stolyarov, S. K. Stoszek, A. von Gottberg, D. Wurzel, L.-M. Yoshida, C. F. Yung, H. J. Zar, Respiratory Virus Global Epidemiology Network, H. Nair, RESCEU investigators, Global, regional, and national disease burden estimates of acute lower respiratory infections due to respiratory syncytial virus in children younger than 5 years in 2019: A systematic analysis. *Lancet* **399**, 2047–2064 (2022).
45. J.-L. Bayart, J. Degosserie, J. Favresse, C. Gillot, M. Didembourg, H. P. Djokoto, V. Verbelen, G. Roussel, C. Maschietto, F. Mullier, J.-M. Dogné, J. Douxfils, Analytical sensitivity of six SARS-CoV-2 rapid antigen tests for omicron versus delta variant. *Viruses* **14**, 654 (2022).
46. M. Bekliz, K. Adea, O. Puhach, F. Perez-Rodriguez, S. M. Melancia, S. Baggio, A.-R. Corvaglia, F. Jacquerioz, C. Alvarez, M. Essaidi-Laziosi, C. Escadafal, L. Kaiser, I. Eckerle, Analytical sensitivity of eight different SARS-CoV-2 antigen-detecting rapid tests for omicron-BA.1 Variant. *Microbiol. Spectr.* **10**, e0085322 (2022).
47. C. E. Brook, G. R. Northrup, A. J. Ehrenberg, IGI SARS-CoV-2 Testing Consortium, J. A. Doudna, M. Boots, Optimizing COVID-19 control with asymptomatic surveillance testing in a university environment. *Epidemics* **37**, 100527 (2021).
48. M. J. Mina, R. Parker, D. B. Larremore, Rethinking COVID-19 test sensitivity—A strategy for containment. *N. Engl. J. Med.* **383**, e120 (2020).
49. I. Torjesen, COVID-19: Peak of viral shedding is later with omicron variant, Japanese data suggest. *BMJ* **376**, o89 (2022).
50. S. Hakki, J. Zhou, J. Jonnerby, A. Singanayagam, J. L. Barnett, K. J. Madon, A. Koycheva, C. Kelly, H. Houston, S. Nevin, J. Fenn, R. Kundu, M. A. Crone, T. D. Pillay, S. Ahmad, N. Derqui-Fernandez, E. Conibear, P. S. Freemont, G. P. Taylor, N. Ferguson, M. Zambon, W. S. Barclay, J. Dunning, A. Lalvani, A. Badhan, R. Varro, C. Luca, V. Quinn, J. Cutajar, N. Nichols, J. Russell, H. Grey, A. Ketkar, G. Miserocchi, C. Tejpal, H. Catchpole, K. Nixon, B. D. Biase, T. Hopewell, J. S. Narean, J. Samuel, K. Timcang, E. M. Dermott, S. Bremang, S. Hammett, S. Evetts, A. Kondratiuk, Onset and window of SARS-CoV-2 infectiousness and temporal correlation with symptom onset: A prospective, longitudinal, community cohort study. *Lancet Respir. Med.* **10**, 1061–1073 (2022).
51. U.S. Centers for Disease Control and Prevention. Isolation and precautions for people with COVID-19. https://cdc.gov/coronavirus/2019-ncov/your-health/isolation.html. [accessed 11 May 2023].
52. D. Larremore, D. Toomre, R. Parker, Modeling the effectiveness of olfactory testing to limit SARS-CoV-2 transmission. *Nat. Commun.* **12**, 3664 (2021).
53. C. Fraser, S. Riley, R. M. Anderson, N. M. Ferguson, Factors that make an infectious disease outbreak controllable. *Proc. Natl. Acad. Sci. U.S.A.* **101**, 6146–6151 (2004).
54. J. Hellewell, S. Abbott, A. Gimma, N. I. Bosse, C. I. Jarvis, T. W. Russell, J. D. Munday, A. J. Kucharski, W. J. Edmunds, Centre for the Mathematical Modelling of Infectious Diseases COVID- Working Group, S. Funk, R. M. Eggo, Feasibility of controlling COVID-19 outbreaks by isolation of cases and contacts. *Lancet Glob Health* **8**, e488–e496 (2020).
55. K. M. Bubar, K. Reinholt, S. M. Kissler, M. Lipsitch, S. Cobey, Y. H. Grad, D. B. Larremore, Model-informed COVID-19 vaccine prioritization strategies by age and serostatus. *Science* **371**, 916–921 (2021).
56. C. Pretorius, J. Stover, L. Bollinger, N. Bacaër, B. Williams, Evaluating the cost-effectiveness of pre-exposure prophylaxis (PrEP) and its impact on HIV-1 transmission in South Africa. *PLOS ONE* **5**, e13646 (2010).
57. K. M. Bubar, C. E. Middleton, K. K. Bjorkman, R. Parker, D. B. Larremore, SARS-CoV-2 transmission and impacts of unvaccinated-only screening in populations of mixed vaccination status. *Nat. Commun.* **13**, 2777 (2022).
58. J. Wallinga, P. Teunis, Different epidemic curves for severe acute respiratory syndrome reveal similar impacts of control measures. *Am. J. Epidemiol.* **160**, 509–516 (2004).
59. J. A. Hay, L. Kennedy-Shaffer, S. Kanjilal, N. J. Lennon, S. B. Gabriel, M. Lipsitch, M. J. Mina, Estimating epidemiologic dynamics from cross-sectional viral load distributions. *Science* **373**, eabh0635 (2021).
60. T. A. Lieu, G. Thomas Ray, N. P. Klein, C. Chung, M. Kulldorff, Geographic clusters in underimmunization and vaccine refusal. *Pediatrics* **135**, 280–289 (2015).

Downloaded from https://www.science.org on March 08, 2026

61. S. M. Kissler, N. Kishore, M. Prabhu, D. Goffman, Y. Beilin, R. Landau, C. Gyamfi-Bannerman, B. T. Bateman, J. Snyder, A. S. Razavi, D. Katz, J. Gal, A. Bianco, J. Stone, D. Larremore, C. O. Buckee, Y. H. Grad, Reductions in commuting mobility correlate with geographic differences in SARS-CoV-2 prevalence in New York City. *Nat. Commun.* **11**, 4674 (2020).
62. D. Holtz, M. Zhao, S. G. Benzell, C. Y. Cao, M. A. Rahimian, J. Yang, J. Allen, A. Collis, A. Moehring, T. Sowrirajan, D. Ghosh, Y. Zhang, P. S. Dhillon, C. Nicolaides, D. Eckles, S. Aral, Interdependence and the cost of uncoordinated responses to COVID-19. *Proc. Natl. Acad. Sci. U.S.A.* **117**, 19837–19843 (2020).
63. T. W. Russell, H. Townsley, S. Abbott, J. Hellewell, E. J. Carr, L. Chapman, R. Pung, B. J. Quilty, D. Hodgson, A. S. Fowler, L. Adams, C. Bailey, H. V. Mears, R. Harvey, B. Clayton, N. O'Reily, Y. Ngai, J. Nicod, S. Gamblin, B. Williams, S. Gandhi, C. Swanton, R. Beale, D. L. Bauer, E. C. Wall, A. Kucharski, Within-host SARS-CoV-2 viral kinetics informed by complex life course exposures reveals different intrinsic properties of omicron and delta variants. medRxiv 2023.05.17.23290105 [Preprint] (2023). https://doi.org/10.1101/2023.05.17.23290105.
64. S. Zhang, P. R. Heck, M. N. Meyer, C. F. Chabris, D. G. Goldstein, J. M. Hofman, An illusion of predictability in scientific results: Even experts confuse inferential uncertainty and outcome variability. *Proc. Natl. Acad. Sci. U.S.A.* **120**, e2302491120 (2023).
65. T. C. Jones, G. Biele, B. Mühlemann, T. Veith, J. Schneider, J. Beheim-Schwarzbach, T. Bleicker, J. Tesch, M. L. Schmidt, L. E. Sander, F. Kurth, P. Menzel, R. Schwarzer, M. Zuchowski, J. Hofmann, A. Krumbholz, A. Stein, A. Edelmann, V. M. Corman, C. Drosten, Estimating infectiousness throughout SARS-CoV-2 infection course. *Science* **373**, eabi5273 (2021).
66. S. E. Landis, V. J. Schoenbach, D. J. Weber, M. Mittal, B. Krishan, K. Lewis, G. G. Koch, Results of a randomized trial of partner notification in cases of HIV infection in North Carolina. *N. Engl. J. Med.* **326**, 101–106 (1992).
67. J. O. Lloyd-Smith, S. J. Schreiber, P. Ekkehard Kopp, W. M. Getz, Superspreading and the effect of individual variation on disease emergence. *Nature* **438**, 355–359 (2005).
68. N. Néant, G. Lingas, Q. L. Hingrat, J. Ghosn, I. Engelmann, Q. Lepiller, A. Gaymard, V. Ferré, C. Hartard, J.-C. Plantier, V. Thibault, J. Marlet, B. Montes, K. Bouiller, F.-X. Lescure, J.-F. Timsit, E. Faure, J. Poissy, C. Chidiac, F. Raffi, A. Kimmoun, M. Etienne, J.-C. Richard, P. Tattevin, D. Garot, V. L. Moing, D. Bachelet, C. Tardivon, X. Duval, Y. Yazdanpanah, F. Mentré, C. Laouénan, B. Visseaux, J. Guedj, French COVID Cohort Investigators, and French Cohort Study groups, Modeling SARS-CoV-2 viral kinetics and association with mortality in hospitalized patients from the French COVID cohort. *Proc. Natl. Acad. Sci. U.S.A.* **118**, e2017962118 (2021).
69. L. M. Kucirka, S. A. Lauer, O. Laeyendecker, D. Boon, J. Lessler, Variation in false-negative rate of reverse transcriptase polymerase chain reaction–based SARS-CoV-2 tests by time since exposure. *Ann. Intern. Med.* **173**, 262–267 (2020).
70. H. M. Dobrovolny, M. B. Reddy, M. A. Kamal, C. R. Rayner, C. A. A. Beauchemin, Assessing mathematical models of influenza infections using features of the immune response. *PLOS ONE* **8**, e57088 (2013).
71. M. Wathuo, G. F. Medley, D. J. Nokes, P. K. Munywoki, Quantification and determinants of the amount of respiratory syncytial virus (rsv) shed using real time PCR data from a longitudinal household study. *Wellcome Open Res.* **1**, 27 (2016).
72. R. Ke, P. P. Martinez, R. L. Smith, L. L. Gibson, C. J. Achenbach, S. M. Fall, C. Qi, J. Jacob, E. Dembele, C. Bundy, L. M. Simons, E. A. Ozer, J. F. Hultquist, R. Lorenzo-Redondo, A. K. Opdycke, C. Hawkins, R. L. Murphy, A. Mirza, M. Conte, N. Gallagher, C. H. Luo, J. Jarrett, A. Conte, R. Zhou, M. Farjo, G. Rendon, C. J. Fields, L. Wang, R. Fredrickson, M. E. Baughman, K. K. Chiu, H. Choi, K. R. Scardina, A. N. Owens, J. Broach, B. Barton, P. Lazar, M. L. Robinson, H. H. Mostafa, Y. C. Manabe, A. Pekosz, D. D. McManus, C. B. Brooke, Longitudinal analysis of SARS-CoV-2 vaccine breakthrough infections reveal limited infectious virus shedding and restricted tissue distribution. *Open Forum Infect. Dis.* **9**, ofac192 (2022).
73. K.-C. Tsaoa, Y.-B. Kuob, C.-G. Huanga, S.-W. Chaue, E.-C. Chan, Performance of rapid-test kits for the detection of the pandemic influenza A/H1N1 virus. *J. Virol. Methods* **173**, 387–389 (2011).
74. QuickVue RSV10 Test. QuickVue RSV10 rapid antigen test product specification. https://quidel.com/sites/default/files/product/documents/EF1179501EN01.pdf.
75. A. Pekosz, V. Parvu, M. Li, J. C. Andrews, Y. C. Manabe, S. Kodsi, D. S. Gary, C. Roger-Dalbert, J. Leitch, C. K. Cooper, Antigen-based testing but not real-time polymerase chain reaction correlates with severe acute respiratory syndrome coronavirus 2 viral culture. *Clin. Infect. Dis.* **73**, e2861–e2866 (2021).
76. E. N. Gallichotte, K. M. Quicke, N. R. Sexton, E. Fitzmeyer, M. C. Young, A. J. Janich, K. Dobos, K. L. Pabilonia, G. Gahm, E. J. Carlton, G. D. Ebel, N. Ehrhart, Early adoption of longitudinal surveillance for SARS-CoV-2 among staff in long-term care facilities: Prevalence, virologic and sequence analysis. *Microbiol. Spectr.* **9**, e0100321 (2021).
77. S. A. Bayat, J. Mundodan, S. Hasnain, M. Sallam, H. Khogali, D. Ali, S. Alateeg, M. Osama, A. Elberdiny, H. Al-Romaihi, M. H. J. Al Thani, Can the cycle threshold (Ct) value of RT-PCR test for SARS CoV2 predict infectivity among close contacts? *J. Infect. Public Health* **14**, 1201–1205 (2021).

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